

AYUSH SINGH

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EDUCATION

Bachelor of Technology in Computer Science with Specialization in Artificial intelligence and Machine Learning | Vellore Institute Of Technology, Bhopal

- August,2023 – May,2027
 - GPA: 8.66
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PROJECTS

Breast Cancer Prediction & Prevention Platform | <https://github.com/AyushSingh331/Breast-Cancer-Prediction-and-Prevention>

- **Model Optimization:** Implemented and fine-tuned an EfficientNetB3 architecture in TensorFlow to perform high-precision medical image classification.
- **Full-Stack Engineering:** Developed a full-stack medical imaging web application using Flask and PostgreSQL, incorporating a dynamic dashboard for comprehensive prediction tracking.
- **Data Security:** Engineered a robust user authentication system utilizing SHA-256 cryptographic hashing to strictly safeguard sensitive patient records and account credentials.

Dog Vision: Multi-Class Breed Identification | <https://github.com/AyushSingh331/Dog-Vision>

- **Deep Learning Pipeline:** Architected a deep learning pipeline using TensorFlow, Pandas and NumPy to ingest and process a massive dataset from the Kaggle Dog Breed Identification competition.
 - **Transfer Learning:** Utilized MobileNetV2 transfer learning to train and test a model capable of differentiating between 120+ unique dog breeds.
 - **Scale & Performance:** Successfully trained the model on over 10,000 images, optimizing accuracy for fine-grained, multi-class classification tasks.
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TECHNICAL SKILLS

- **Languages:** Python, Java, SQL
- **Frameworks & Libraries:** TensorFlow, PyTorch, Scikit-Learn, Flask
- **Data Science:** NumPy, Pandas, Matplotlib
- **Tools & Databases:** PostgreSQL, Git, GitHub